

Health and Wellness of US College Students

Final Report

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Introduction:

The health and wellbeing of college students in the U.S. has become a growing topic of interest for educators, university administration and students themselves. Many students can face high levels of stress as they try to excel academically, whilst also balancing social lives, juggling extracurricular commitments, and maintaining healthy lifestyles. As a result, these students might develop poor habits as a response to this pressure.

For this project, we used two datasets from Kaggle - the *Student Mental Health Survey* dataset by Saint and the *Student Health Data* dataset by Ziya in order to explore the relationship between students' mental and physical health. Our goals are to identify trends between health and academic performance, and to examine whether we can predict how stressed (or relaxed) a student feels based on various factors such as heart rate, sleep quality, and study habits.

Data:

The two datasets that we are analysing for this project were published separately, by the users Saint and Ziya; both on the website Kaggle.

The main dataset, *Student Mental Health Survey* published by Saint, contains survey results from college students collected around 2020-21. Collected through various anonymous resources, this dataset represents mental health evaluations of students and seeks to provide valuable insights into the mental health of students. This survey attempted to capture a number of factors that may impact students' mental health. Omitting any entries with missing or blank data, the sample size of this dataset is $N = 6,973$. A brief summary of the metrics in this dataset are shown below:

- Age: $\{x: 18 \leq x \leq 35, x \in \mathbb{C}\}$.
- Major: $\{\text{Law, Medical, Engineering, Computer Science, Business}\}$.
- Gender: $\{\text{Female, Male}\}$.
- Cumulative GPA (CGPA): $\{x: 2.44 \leq x \leq 4.00, x \in \mathbb{Z}\}$.
- Stress Level: level self-reported by students $\{x: 0.00 \leq x \leq 5.00, x \in \mathbb{Z}\}$.
- Depression Score: score self-reported by students $\{x: 0.00 \leq x \leq 5.00, x \in \mathbb{Z}\}$.
- Anxiety Score: score self-reported by students $\{x: 0.00 \leq x \leq 5.00, x \in \mathbb{Z}\}$.
- Sleep Quality: $\{\text{Poor, Average, Good}\}$.
- Physical Activity: $\{\text{Low, Moderate, High}\}$.
- Diet Quality: $\{\text{Poor, Average, Good}\}$.
- Social Support: $\{\text{Low, Moderate, High}\}$.
- Relationship Status: $\{\text{Married, In a relationship, Single}\}$.
- Substance Use: $\{\text{Frequently, Occasionally, Never}\}$.
- Family History: Whether the individual has a family history of mental health issues $\{\text{Yes, No}\}$.

- Chronic Illness: {Yes, No}.
- Financial Stress: The level of financial stress experienced by the student $\{x: 0.00 \leq x \leq 5.00, x \in \mathbb{Z}\}$.
- Counseling Service Use: {Frequently, Occasionally, Never}.
- Extracurricular Involvement: {Low, Moderate, High}.
- Semester Credit Load: The number of credits students take in the semester $\{x: 15 \leq x \leq 30, x \in \mathbb{Q}\}$.
- Residence Type: {On-campus, Off-campus, With family}.

The *Student Health Data* dataset contains simulated physiological, psychological, and academic data for college students. The dataset focuses on assessing health risk due to high-stress environments like academic and entrepreneurial settings. It was created to support the development and testing of machine learning models that predict occupational health risks, such as stress and physical strain. The data includes both sensor-based measurements and self-reported values. The sample size of this dataset is $N = 1,000$ with no missing data.

- Age: $\{x: 18 \leq x \leq 24, x \in \mathbb{Z}\}$.
- Gender: {Female, Male}.
- Heart Rate: {BPM}.
- Blood Pressure (Systolic, Diastolic): {mmHg}.
- Self-reported Stress Levels: $\{x: 1.00 \leq x \leq 10.00, x \in \mathbb{Q}\}$.
- Biosensor Stress Level: $\{x: 1.00 \leq x \leq 10.00, x \in \mathbb{Q}\}$.
- Mood States: {Stressed, Neutral, Happy}.
- Academic Activity: {hours}
- Entrepreneurial Activity: {hours}
- Daily Physical Activity: {hours}
- Sleep Quality: {Poor, Moderate, Good}
- Health Risk Level: {Low, Moderate, High}.

Research Questions:

1. Can we find an approximate model for the distribution of GPA and resting heart rate among students and what are the estimates of its parameters (parameter estimation)?
2. Can we build a good linear model that predicts a student's self-reported stress levels given their biosensor stress report or their heart rate, study hours, physical activity or sleep quality?
3. Are there significant differences between different groups of students and their average mental health, physical health, or GPA and what would be the results of a t-test comparing CGPA and Sleep Deprivation in students (hypothesis testing)?

4. Does mean self-reported stress differ significantly among students reporting High, Moderate, and Low physical-activity levels in the Saint Student Mental Health dataset? Conduct a one-way ANOVA to test for group differences.
5. Is there a good regression model that predicts a student's stress level, depression score, or anxiety level based on predictors including family history, chronic illness, financial stress, semester credit load, and resident type (regression model)? Would this model be more accurate with other parameters (major, gender, self-reported social support score)?
6. Can we build a good linear model that predicts a student's cumulative GPA given their mental and physical health (WLS regression)?

Section 1: Distribution Modeling and Parameter Estimation

Contained in the Student Mental Health Assessment dataset are the cumulative GPAs of each student surveyed. A student's GPA can be used as a measurement of their academic performance, so before comparing across each group, it's important to find an approximate model of this data. It's important to note that this metric is *cumulative*, meaning some caution should be taken when comparing students who have just started taking classes - typically younger - who may not have a fully representative GPA compared to students who are nearing their graduation - typically older.

Simple statistics tell us that the mean of this data is 3.49, the variation is 0.083, and the data ranges from 2.44 to 4.0. The most notable thing about the distribution of this data is the large spike at the value '4.0' compared to the rest of the distribution, which makes it difficult to fit a model. This is most likely a result of the ceiling effect, where 4.0 is the maximum possible grade-point average and rounding parameters perhaps weren't specified in the collection method.

The data is depicted in a histogram below (Figure 1.1). We've fitted a normal, bimodal, KDE, clustering model, and beta density distributions. Visually, it appears that the kernel density and clustering model distributions fit this data best, because they both account for the large spike at 4.0. For further interpretation, however, we have chosen to use the clustering model, as KDE won't be a helpful tool in analysis. This clustering model was created through the Mclust package, which is a Gaussian finite mixture model tool that uses EM algorithm. **This Mclust model determines specified bins through its EM algorithm, which uses maximum likelihood estimation to partition these hierarchical clusters.**

A summary of all the models and their insights into this distribution is shown below in Table 1.1. We've included all of these metrics because they're useful insights into the spread of this distribution, such as the unequal variances and two separate modes (and subsequent means).

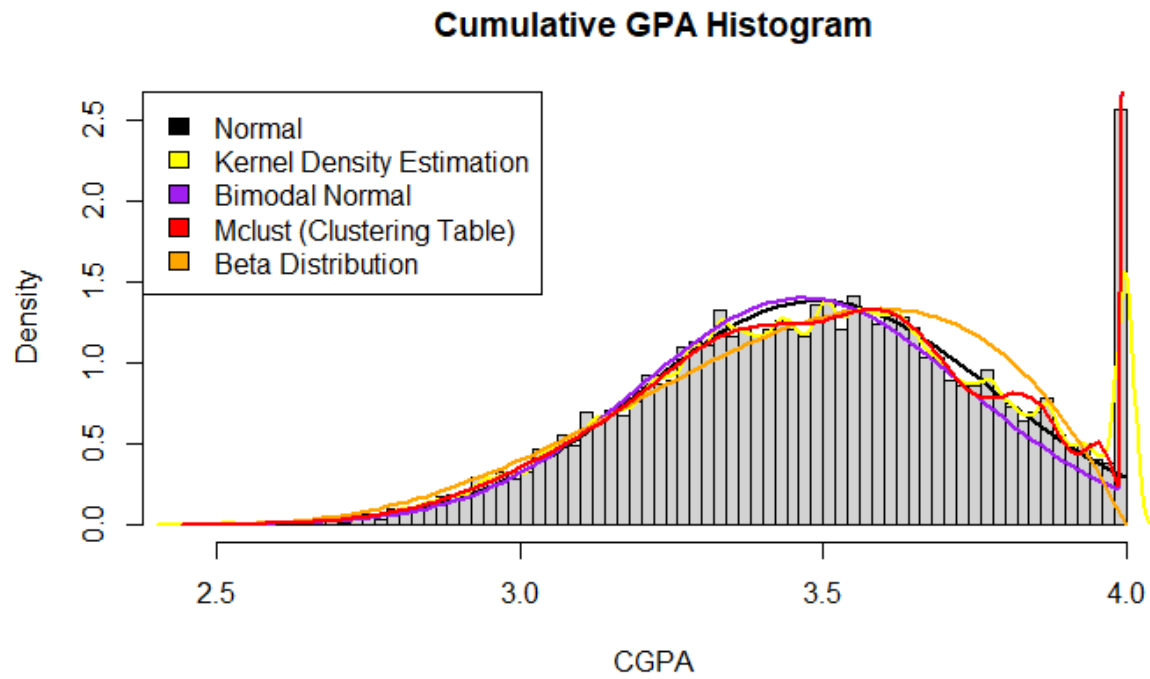


Figure 1.1: Distribution of cumulative GPA in this sample.

Model						
Normal	$\mu = 3.491172$			$\sigma = 0.287595$		
Bimodal	$\mu_1 = 3.466034$	$\mu_2 = 3.999996810$	$\sigma_1 = 0.270856$	$\sigma_2 = 0.0005967$	$\lambda_1 = 0.952936$	$\lambda_2 = 0.047063641$
Kernel Density Estimation (KDE)	5th quantile ≈ 3.009	25th quantile ≈ 3.29	50th quantile ≈ 3.5	75th quantile ≈ 3.7	95th quantile ≈ 3.979	100th quantile ≈ 4.00
Clustering Mclust Model (cluster means & variances)	Clustering Table (counts in each bin)					
	2185	1140	2498	645	173	332
	3.240640	3.385582	3.629229	3.848417	3.958333	3.999949
	0.0488316	0.01467678	0.01282868	0.003627482	0.0005277885	3.253409e-07

Table 1.1: Summarized information for each estimator (aside from the beta distribution).

We also ran parameter estimation to discover if we could estimate the distribution of students resting heart rate. Figure 1.2 shows the distribution of resting heart rates in Ziya's Student Health Data, forming a classic bell curve with most students clustered around 70 bpm. When we fit a Normal distribution to these measurements, we estimate the mean to be $\hat{\mu} = 70.10$ bpm (95 % CI [69.51, 70.69]) and the standard deviation to be $\hat{\sigma} = 9.54$ bpm (95 % CI [9.14, 9.97]). Because these values fall almost exactly within the 60 to 80 bpm range reported for healthy young adults, we can be confident that our sample behaves like a typical college population. The tight confidence intervals also mean that this Normal approximation will serve reliably in downstream analyses such as simulating heart rate scenarios or feeding values into stress prediction models.

To capture any slight skew and to ensure non negative values, we also overlaid a two parameter Gamma fit in Figure 1.2. That model yields a shape parameter $\hat{k} = 53.29$ (95 % CI [49.70, 58.58]) and a scale parameter $\hat{\theta} = 1.315$ bpm (95 % CI [1.20, 1.41]). Its implied mean, calculated as k times θ , is approximately 70 bpm and its variance closely matches the Normal fit, which means either distribution could underpin future simulation studies. In practice, the Gamma model's slight skew toward higher values is especially useful when heart rate spikes occur (for example, after sprinting across campus) while automatically ruling out impossible negative measurements.

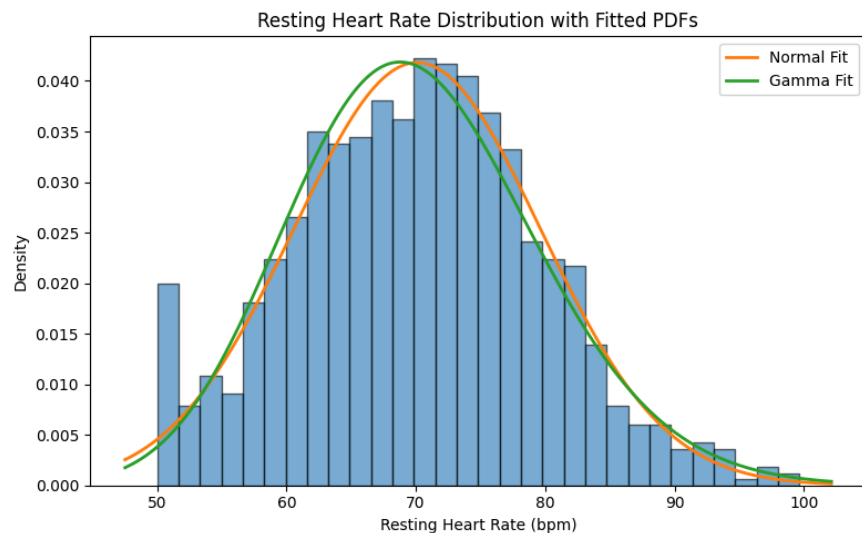


Figure 1.2: Resting Heart Rate Distribution with Fitted PDFs

Section 2: Stress Predictor Linear Regression Model

In this section, we built a multiple linear regression model to predict how stressed a student might self-report as, using the predictor variables of sleep quality (categorical), mood (categorical), time spent studying (numeric), biosensor-reported stress level (numeric), heart rate (numeric), physical activity (categorical), and health risks (categorical). We chose this question because understanding which factors best predict a student's perceived stress could provide students, university administrators, and caregivers with valuable insight into the contributors of student stress. Additionally, by including biosensor-reported stress as a predictor, we can explore how closely physiological stress markers align with a student's self-perception of stress - do students feel as stressed, more stressed, or less stressed than their biological signals suggest?

To visualise the results of the multiple linear regression model, we used effect plots. These plots show the predicted values of perceived stress based on each individual predictor (e.g., heart rate, or sleep quality), while holding the other variables constant. This approach helps isolate the unique contribution of each factor to self-reported stress, providing a clearer understanding of which variables have the strongest relationship with perceived stress in students. The y-axis reflects the model's predicted stress values, which is why the range appears narrower than the original 1–10 scale used in the data. These effect plots were created using the `effects` package.

Initially, we created a model using heart rate, mood, sleep quality, biosensor stress report, and study hours as the predictor variables (shown in Figure 2.1). The results showed that this model explained approximately 0.39% of the variance in self-reported stress levels (Adjusted $R^2 = -0.003145$), and they indicated that none of the predictor variables were statistically significant with all p-values exceeding 0.3.

These results suggest that, within this model, the included factors – while related to stress in some capacity- were not strong predictors of how stressed a student reported feeling in this dataset. Although this model has low predictive power, it highlights the complexity of stress and suggests that self-reported stress may be influenced by factors beyond the variables we chose to focus on.

To explore whether a more predictive model could be built, a second multiple linear regression model was created (Figure 2.2) using the same `effects` package to visualise the effects of each predictor on self-reported stress. However, this time we used a new set of predictor variables: physical activity, mood, study hours, health risks and sleep quality.

In contrast, the results showed that this second model explained approximately **26.0%** of the variance in self-reported stress levels (Adjusted $R^2 = 0.2602$), indicating a substantial

improvement in the explanatory power compared to the first model. Additionally, there were several statistically significant variables:

- Low Physical Activity and Moderate Physical activity both had p-values < 0.001
- Low Health Risk and Moderate Health Risk also had p-values < 0.001

Yet, study hours, mood, and sleep quality were not statistically significant in this model (p-value > 0.05), although sleep quality was somewhat significant with p-value = 0.084. This second model revealed that physical activity and health risk level are key predictors of how stressed a student *feels*. It supports the idea that students with lower physical activity levels and lower health risk scores report higher levels of perceived stress. Even though some predictors like mood and sleep quality might feel relevant when you think about stress without looking at the data, they did not show a significant statistical relationship. The multiple linear regression model is of the form :

$$\hat{Y} = 6.85 + 0.86(PhysicalActivity_{low}) + 0.84(PhysicalActivity_{moderate}) + 0.0027(StudyHours) - 0.08(Sleep_{moderate}) - 0.32(Sleep_{poor}) + 0.02(Mood_{neutral}) - 0.03(Mood_{stressed}) - 4.60(HealthRisk_{low}) - 1.9(HealthRisk_{moderate})$$

, where $\hat{Y}$ is the predicted self-reported stress level

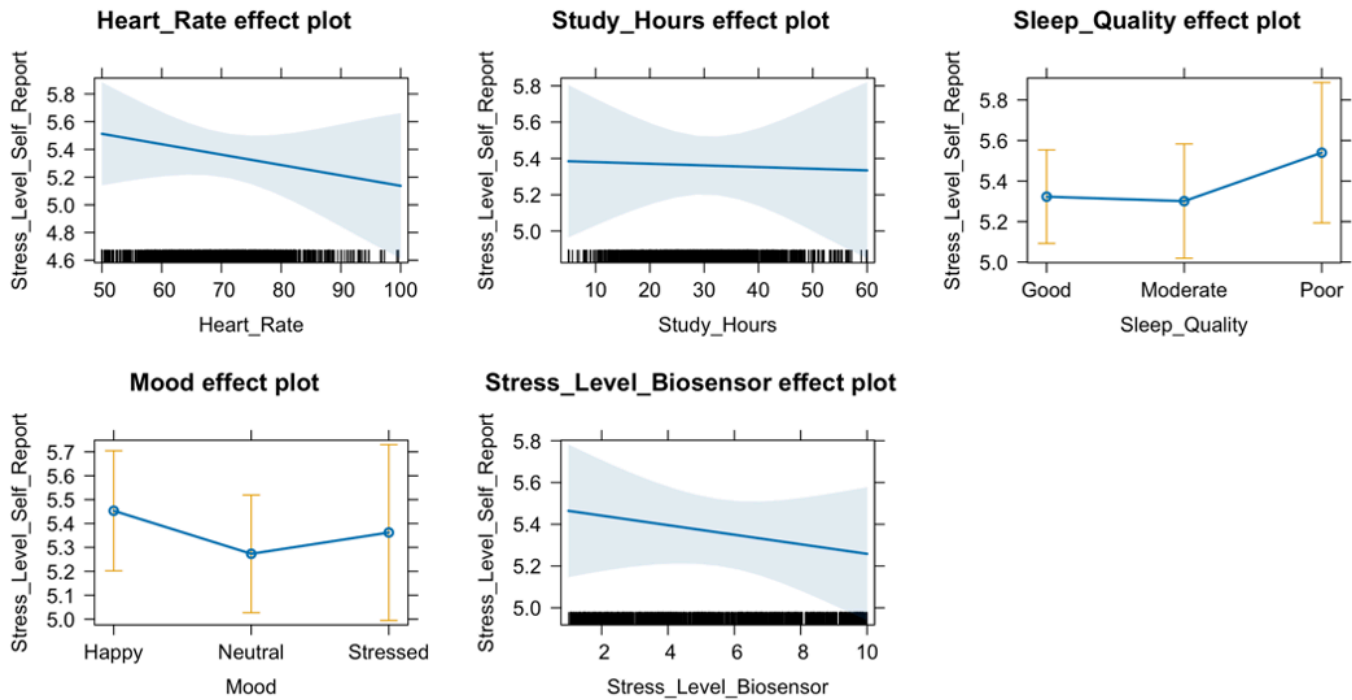


Figure 2.1: Effect Plots for Initial Regression Model Predicting Self-Reported Stress

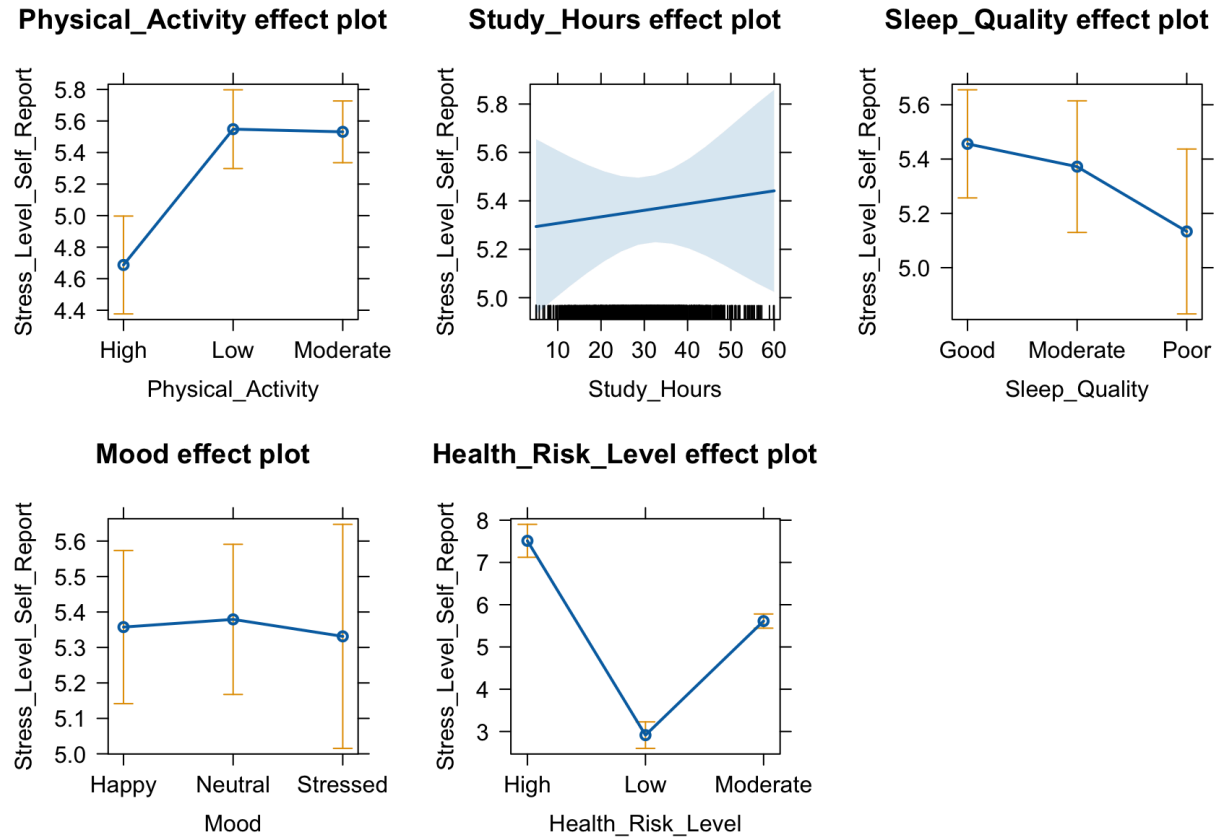


Figure 2.2: Effect Plots for Revised Regression Model with Health and Activity Predictors

Section 3: Hypothesis Testing using various methods

In this section, we use various hypothesis testing methods to discover whether certain groups have significant differences between mental health, physical health, and GPA. All tests are done using the Student Mental Health Assessment dataset.

Mental Health

The Student Mental Health Assessment dataset collects the students perceived stress, anxiety, and depression levels on an integer scale of 0 being none and 5 being very perceived/prevalent. We combined these scores to create a cumulative "mental health score", thus scores that are higher indicate worse mental health. These cumulative scores approximately follow a normal distribution with $\mu = 6.98695$ and $\sigma^2 = 7.455796$. This dataset contains upwards of 6,000 entries, so for this section, we've determined μ and σ^2 through the estimators

$\hat{\mu} = \bar{Y}$ and $\hat{\sigma}^2 = S^2$. With this data approximately following a normal distribution, we have used the Z test statistic to determine p-values for each test.

A full catalogue of fourteen tests can be seen in the attached code report, but to summarize, all of these tests had insignificant results. The greatest significance was testing whether students who have “low” social support have worse mental health compared to students with “moderate” or “high” social support. As defined by the way we conducted this test, a student with “worse” mental health would be defined by a higher “mental health score,” so this hypothesis test compared the mean of the first population, characterised as “low” social support, to the mean of the second population which was labeled with individuals who had “moderate” or “high” social support, to see whether the mean score of the “low” support first population was significantly greater than the second population. With $p\text{-val} = 0.05857$, this test had the smallest p-value of all of the tests run in this section, but still greater than the literature-acceptable significance value of $\alpha = 0.05$. Even though this p-value is greater than 0.05, there’s still some weight to this test, showing that social support is a contributing factor when it comes to the mental health of students.

Physical Health

The Student Mental Health Assessment dataset also collects the conditions of students' sleep quality, diet quality, and physical activity. Because these variables are qualitative, we have utilized analysis of categorical data for this subsection. Meaning, these tests are done using chi-squared test statistics of categorical data. We again summarize the different categorical values students could select for these parameters below:

- Sleep and Diet Quality: "poor", "average", and "good".
- Physical Activity: "low", "moderate", and "high".

We have completed chi-square tests between male and female students and across the different majors supplied by this dataset to discover whether sleep quality, diet quality, and physical activity significantly differs between these groups. After running these tests, the p-values were all insignificant, indicating that there is no significant difference of these parameters between male and female students or across students in different majors.

Although these tests were all insignificant, with p-values all greater than 0.45, there is an interesting metric we came across while running a chi-square test of diet quality across majors. The residuals table is supplied below, which shows a larger than normal value of 2.744 under the “poor” diet quality for computer science majors. While the overall p-value of this test was 0.4849, this one particular value is interesting because it falls outside of the normal assumption for residuals and suggests that more computer science majors have “poor” diet quality than expected. No doubt is this one value contributing to the decreased p-value, but because all of the other residuals fall within normal assumptions, we can’t say much across majors for diet quality.

Data = diet_matrix				
$\chi^2 = 17.561$		df = 18	p-value = 0.4849	
	"Poor"	"Average"	"Good"	TOTAL
Business	-0.9047130	0.44905427	0.10975552	0
Compsci	2.7440270	-1.58055411	0.05080464	0
Engineering	-1.0930620	-0.05977414	1.19002849	0
Law	-0.8740407	0.59580497	-0.17832823	0
Medical	0.7590821	-0.27264755	-0.27488600	0
Others	-0.6016607	0.87466261	-0.93828321	0
TOTAL	0.0000000	0.0000000	0.0000000	0

Table 3.1: Pearson's Chi-Square test analyzing categorical markers for diet quality across different majors. Interesting note is the residual for computer science majors for "poor" diet quality. **EDIT: deleted sentence; there isn't a normal assumption for categorical data.**

GPA

After running models to see the best fitting in Section 1, we've used the Mclust Clustering Model to perform some analysis of GPA across groups. This model combines a similar GPA into 6 bins. Thus, we have employed the use of chi-square tests to discover whether there is any difference between groups using these 6 bins as markers. The summary of the bins are included in Table 1.1.

Tests were run across every metric provided in the dataset (besides age) with all of them returning insignificant. Nonetheless, there are some notable observations in the subsequent residuals table. For instance, we ran the test of GPA "clusters" against a student's credit load, resulting in the lowest p-value of any test at 0.1111. While this p-value isn't significant, the resultant residuals table tells us that students who have a "high" credit load (defined as 24-26 credits) are found less in the GPA cluster "4" than what's expected, with a residual of -2.2851. This cluster has a mean of 3.848 and represents the cluster with the third highest GPA. Any difference from expectancy is note-worthy, but this specific finding isn't particularly helpful because it's not paired with any significant findings or underrepresented groups for "very high" credit load. Additionally, there was another residual that stood out in our test of GPA clusters and

diet quality. With a residual value of 2.3188 and an overall p-value of 0.1559, students with “poor” diet quality are overrepresented in the GPA cluster “4” than what’s expected. Again, this residual, while note-worthy, isn’t particularly insightful and more tests need to be run to actually discover why this residual is higher than expected.

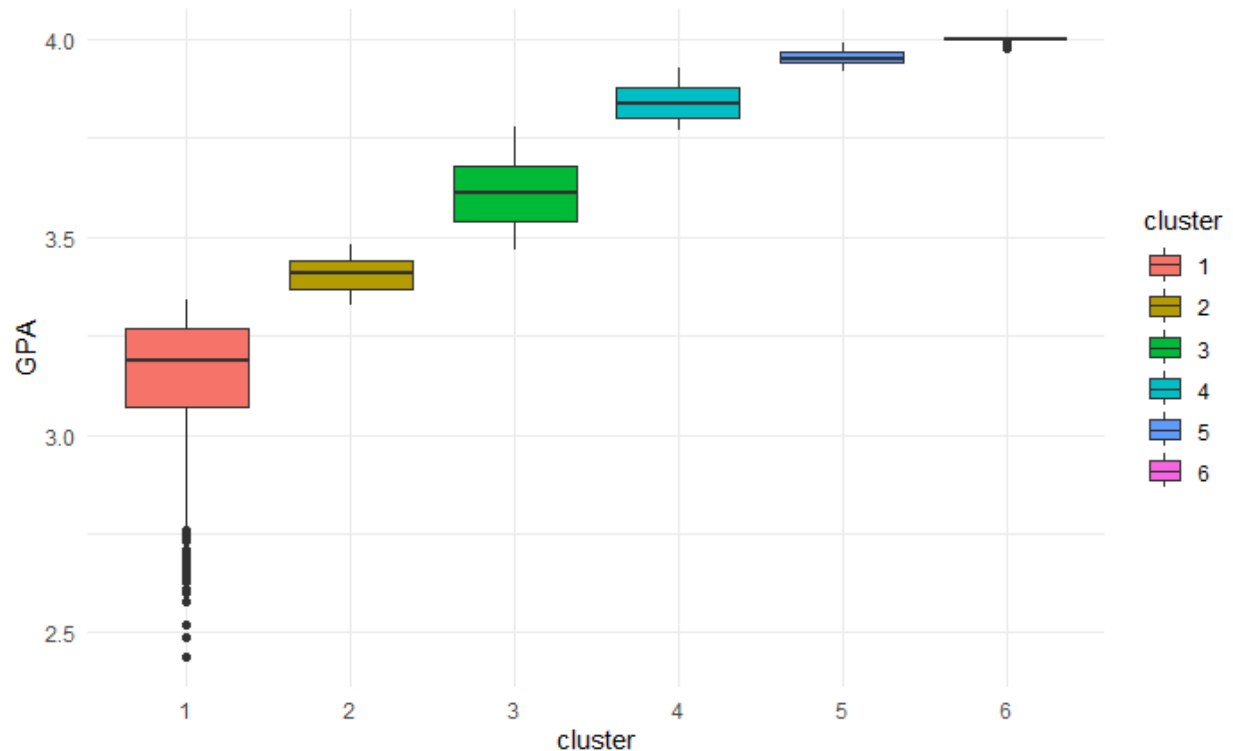


Figure 3.1: Mclust Model Clusters showing the 6 bins used for chi-square analysis. The unequal variances are seen through the height of each box. Algorithm determines how to partition data through maximum likelihood estimation.

Part of what was found in the Kaggle data were cumulative GPA (CGPAs) of students surveyed with their self-reported sleep score (poor, average, and good). To see if sleep deprivation has a relationship with academic performance (CGPA) based on this dataset, a two-sample Welch t test was completed. The outcome of the two-sample t-test ($t = -0.20408$, $p = 0.839$) concludes that there is no meaningful difference between GPA and sleep quality.

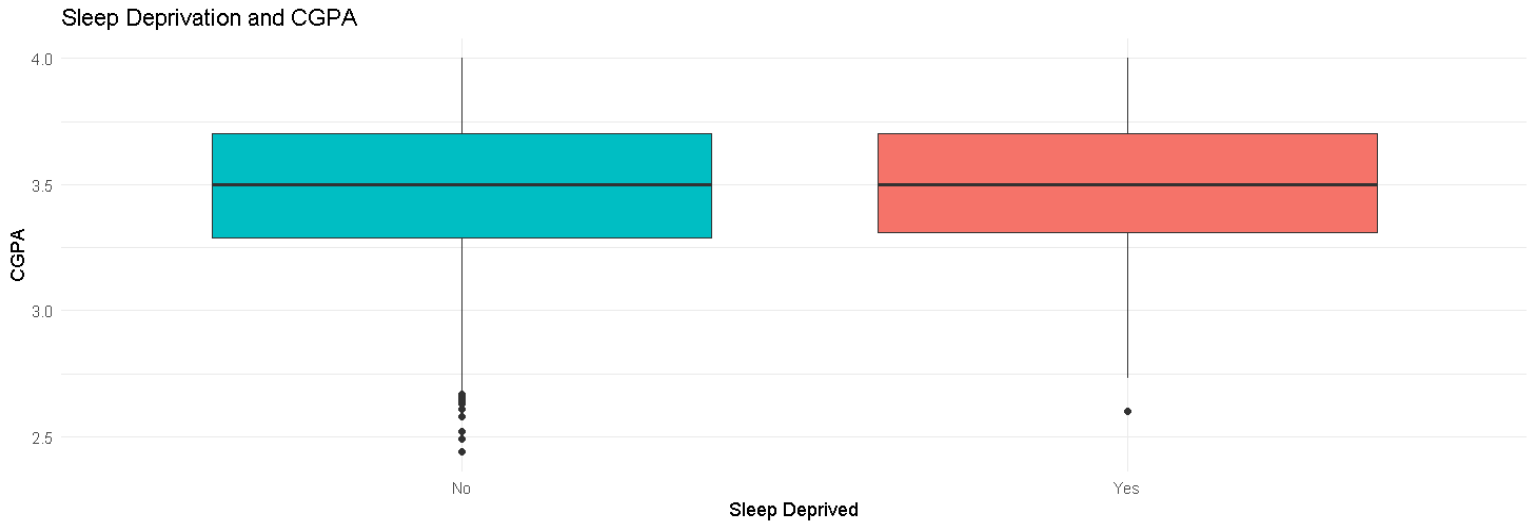


Figure 3.2: : Boxplot comparing Sleep Deprivation and CGPA showing no meaningful difference.

Sleep Deprivation-CGPA T-test	
<i>T-statistic</i>	-0.20408
<i>Degrees of Freedom (df)</i>	60.599
<i>p-value</i>	0.839
<i>95% CI (lower)</i>	-0.0937
<i>95% CI (upper)</i>	0.0763
<i>Mean (Sleep Deprived)</i>	3.4912
<i>Mean (Rested)</i>	3.4999

Table 3.3: T-Test Results for Sleep Deprivation and CGPA.

Section 4: One Way ANOVA

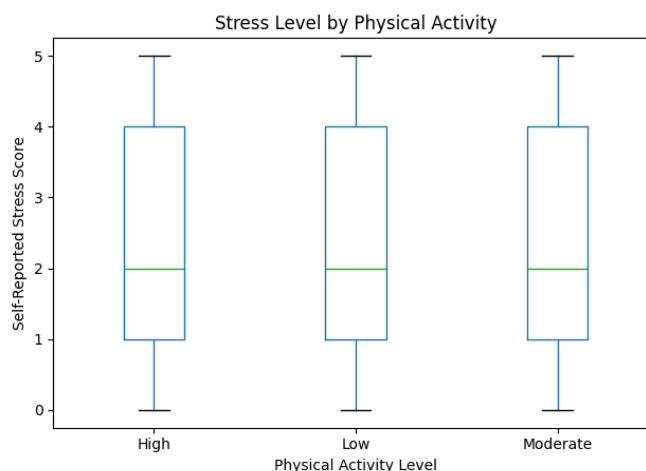


Figure 4.1: Stress Level by Physical Activity

As shown in Figure 4.1, the boxplots of self-reported stress scores for the “High,” “Moderate,” and “Low” physical-activity groups overlap almost perfectly. To quantify this, we framed our **null hypothesis** (H_0) as “the mean stress score is the same in all three activity groups ($\mu_1 = \mu_2 = \mu_3$).” We then ran a one-way ANOVA, which computes an F-statistic by taking the ratio of between-group variance to within-group variance—under H_0 this ratio follows an F-distribution. Our analysis yielded $F = 0.25$ with $p = 0.78$, meaning there’s a 78 % probability of seeing an F-value this large (or larger) if the true group means were equal. Because $p \gg 0.05$, we **fail to reject H_0** , concluding that average stress does not differ by activity level.

This null result aligns with recent campus-wellness surveys suggesting that more exercise alone doesn’t automatically translate into lower perceived stress when academic and social pressures are high. For student-support teams, the takeaway is clear: promoting physical activity remains important for overall health, but it shouldn’t stand alone as a stress-reduction strategy. Pairing fitness programs with mindfulness workshops, time-management coaching, or peer-support groups may be far more effective at helping students manage the real sources of their stress.

Section 5: Student Mental Health Regression Models

In the Student Mental Health Survey dataset, students self-reported mental health scores for depression, anxiety, and stress experienced (scored from 0-5). To research which other parameters of the survey dataset have statistically significant impact on student's mental health, various multiple linear regression and later, ordinal logistic regression models were generated using the same dataset.

One of the goals is to see which parameters of our dataset are most important for understanding student's mental health scores, or which at least lead to the most accurate regression model for predicting student's mental health comparatively, and so the parameters for each regression model vary.

A multiple linear regression model for reported stress (lm model 3) was obtained by simply fitting the data using the R function `lm` given the predictor variables of GPA (numerical), depression score (ordinal), anxiety score (ordinal), and counseling service use (categorical) by the student. The R^2 of this model is 0.0065, with an adjusted R^2 of 0.0047, indicating that, especially after accounting for model complexity, the chosen predictors explain less than 0.5% of the observed variance in stress levels between students.

Another multiple linear regression model for reported stress (lm model 4) was obtained with the same methods, but with different predictor variables: course (categorical) and chronic illness (categorical). This was the only linear regression model in this section with an improved R^2 , of 0.01 (adjusted R^2 also ~ 0.01). Similarly, for comparing CGPA and stress factors for students, the graphs show disconnected data due to the nature of the scores/level being discrete instead of continuous, but later is shown as a demonstration of its limitations as a model.

The two predictors, course and chronic illness, were ultimately found to be the strongest linear predictors identified among all models tested. These two predictors together explain 1% of variance in stress.

It is important to note that Figure 5.1 shows that lm model 4 does not fit the data well, and the standard assumptions for a linear regression model are obviously violated. The Residuals vs Fitted graph shows that the assumption of homoscedasticity is violated by the data (nonlinearity). The Normal Q-Q plot, with residuals deviating at both tails implies non-normal residuals. The data could be skewed rather than normally distributed, or have outliers, though the Residuals vs Leverage plot shows there is only one non-significant outlier.

These diagnostic results for the linear regression model are in line with attempting to fit predictor variables for an ordinal variable such as stress level. Overall, linear regression is not the most appropriate choice for analyzing stress for this particular dataset, though it still gives us some insights into which predictor variables may be most significant for stress. Additionally,

even values such as R^2 may be misleading as all models violate key assumptions. Comparisons based on these statistics should be interpreted with caution.

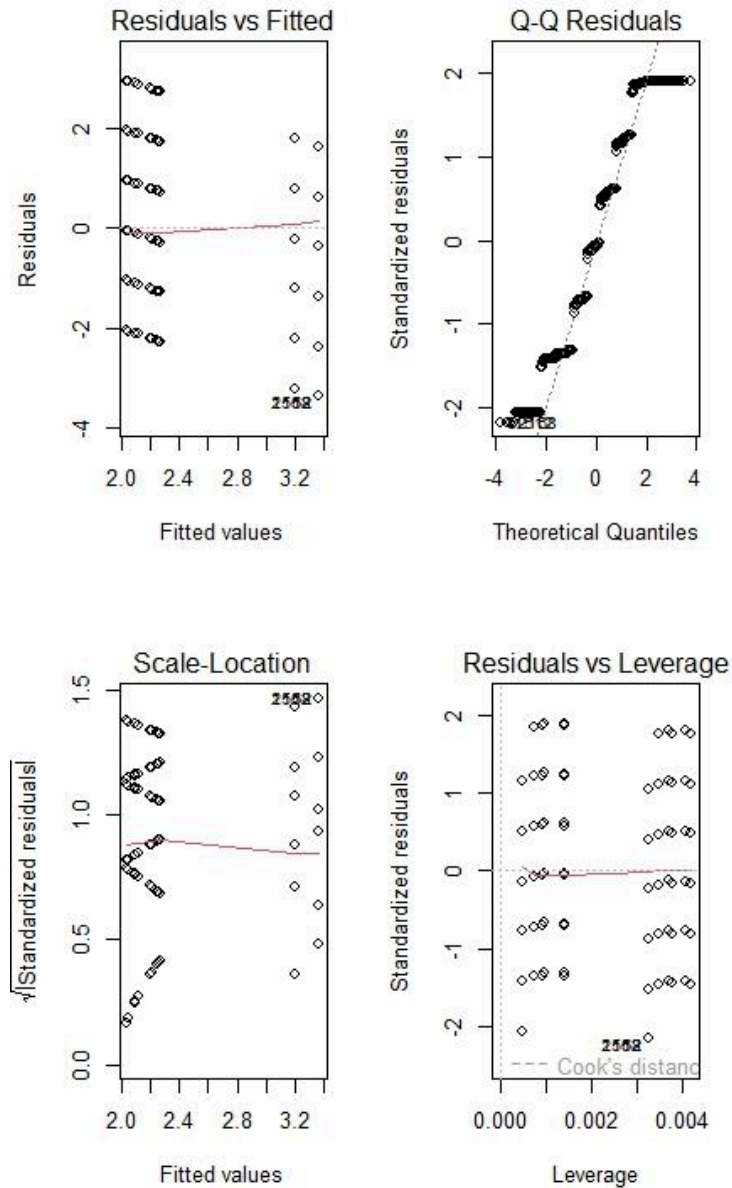


Figure 5.1: Linear Regression Model 3 (Residuals vs Fitted)

Given these issues with performing linear regression, we next give the best found ordinal logistic regression models for the Student Mental Health Survey dataset, as these models are more appropriate for modelling our ordinal outcomes such as reported stress levels.

An ordinal logistic regression model for reported stress (polr model 4) was obtained by fitting the data using the R function `polr` given the predictor variables of course (categorical) and chronic illness (categorical), the same variables as `lm` model 4. Although R^2 values are not defined for logistic regression models, we can still estimate a ‘pseudo- R^2 ’ to help estimate model fit. The pseudo- R^2 of this model is nearly 0.03, which is the highest R^2 found between any logistic models performed. This R^2 was found using the log-likelihood ratio between our full (fitted) model and a null model. This value indicates that the chosen predictors may explain less than 3% of the observed variance in stress levels between students.

Despite the marginal improvement in the explanation of the data’s variance, the ordinal logistic regression results do reveal statistically significant predictors. Notably, the medical course parameter shows a strong positive association with higher stress levels, with a highly significant coefficient ($p < 2e-16$), suggesting this group experiences markedly higher stress compared to the baseline course category. Chronic illness is also a statistically significant predictor, with $p \sim 0.046$. However, the remaining course categories exhibit no statistically significant relationship with reported stress. These results indicate that while the overall model fit may be weak, certain predictors were found to be meaningfully associated with reported stress levels, and all other predictors seem to have limited predictive value of student’s mental health.

Based on this logistic regression model, Figure 5.2 was generated, by using the results of `predict` and the logistic regression model to find the predicted probability for each reported student stress level. Figure 5.2 shows the differences in predicted student stress levels according to our model (student course and chronic illness status). The figure demonstrates that, according to our dataset, students with medical courses are predicted to be far more likely to report higher stress levels. Additionally, Students with chronic illness are slightly higher to report stress scores greater than zero.

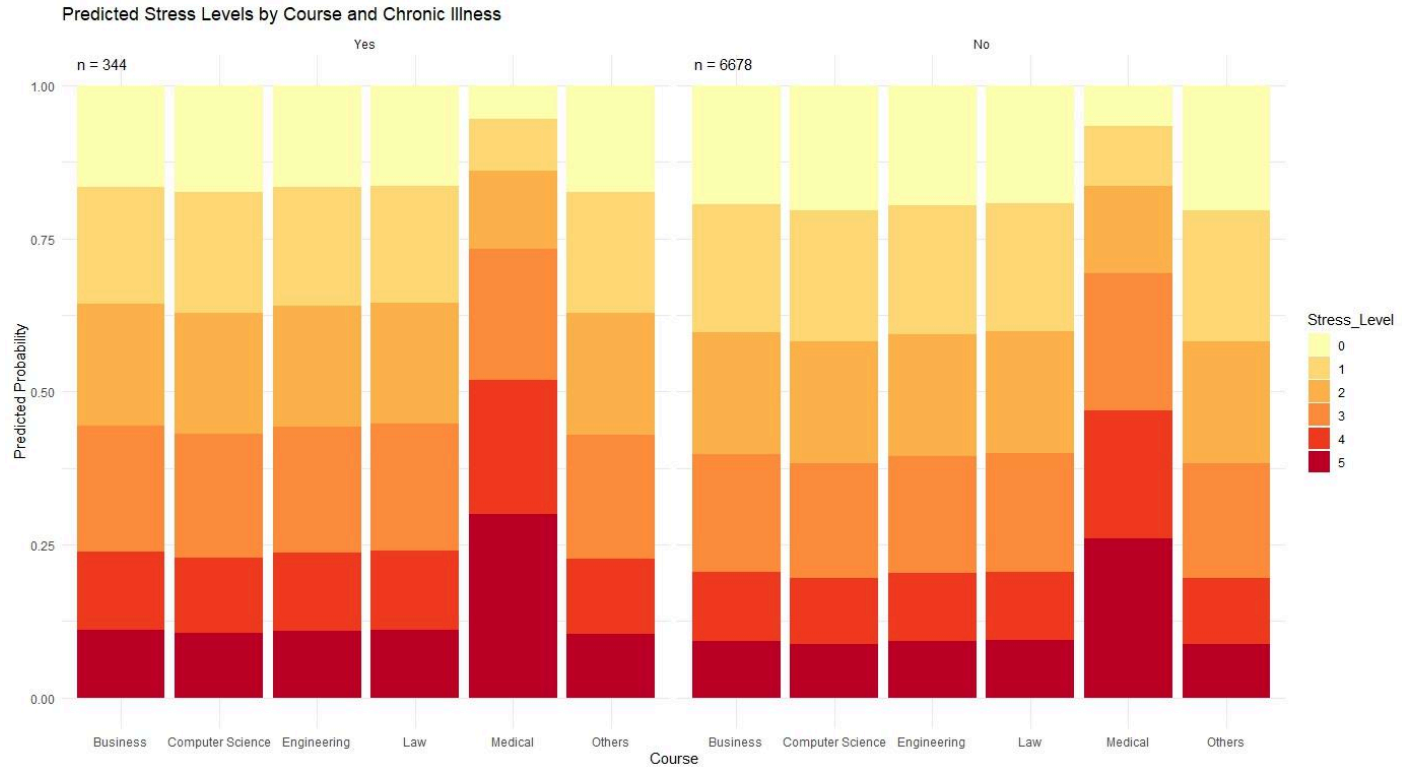


Figure 5.2: Analysis generated from ordinal logistic regression model 4, predicting the probability of all stress level values between students of different courses, and if the student is chronically ill.

In comparison to using multiple linear regression models, it is not as good or helpful in determining if the stress factors are related to CGPA. Because of the limitations of the self-reported scores for stress, anxiety being discrete instead of continuous, it produces horizontal lines in the data and nothing statistically meaningful. The ordinal regression model handles these scores better, but the overall conclusion that can be made about the results is that there is no meaningful correlation between these designated stress factors and what impacts students CGPA from this dataset.

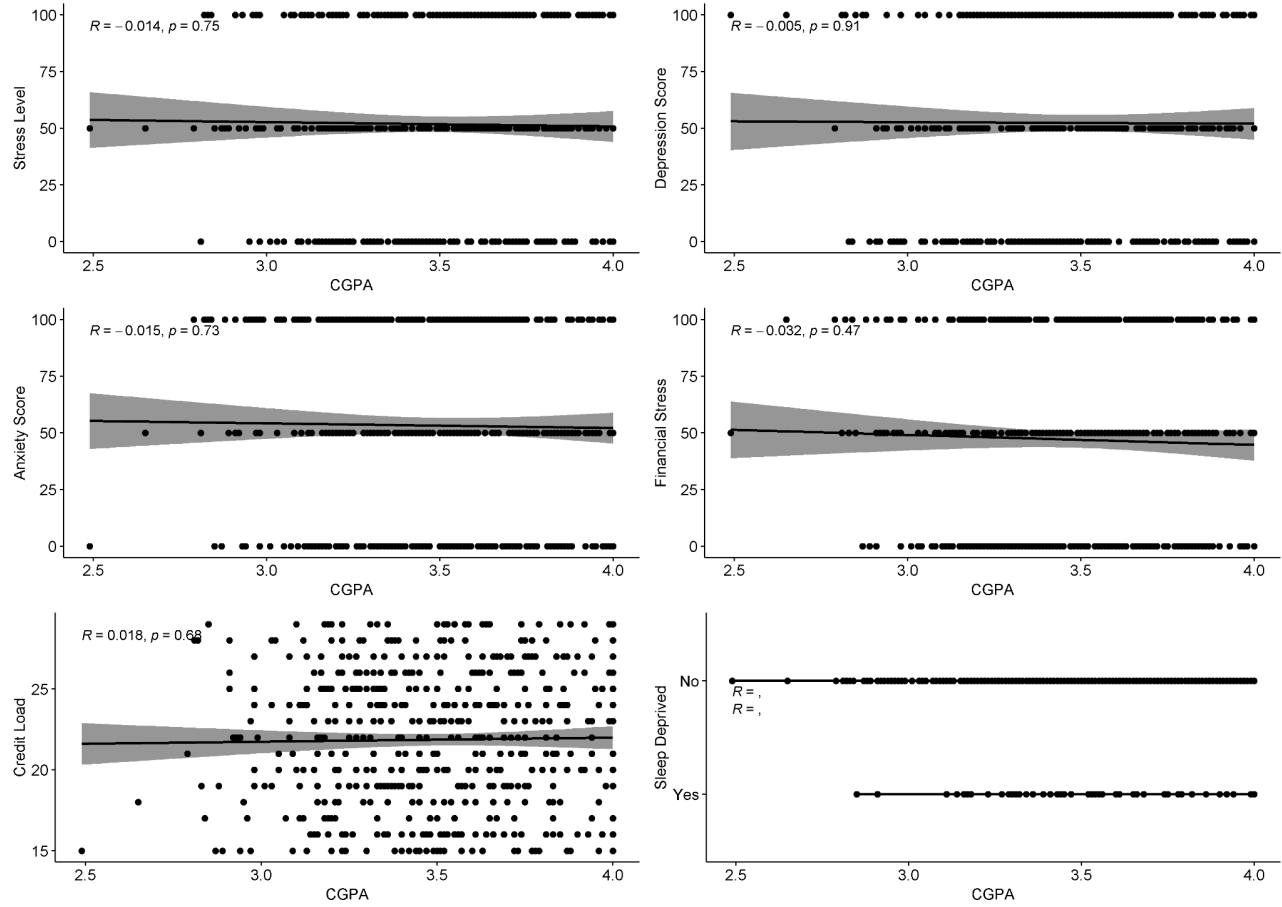


Figure 5.3: : Multiple Linear Regression of Stress Factors comparing and CGPA.

Standardized Regression/ Beta Coefficients	
Sleep Deprivation	0.01003
Stress Level	-0.01429
Depression Score	-0.00688
Anxiety Score	-0.01335
Financial Stress	-0.03159
Semester Credit Load	0.01950

Table 5.1: Beta Coefficient Summary of Stress Factors, showcasing weak effect on CGPA.

EDIT: two whole new sections

Section 6: Predicting CGPA given Mental & Physical Health

In this section, we look to find whether the mental and physical health of a student influences their GPA. Using the factors of sleep quality, diet quality, physical activity, stress level, depression score, and anxiety score in the *Student Mental Health Assessment* dataset, we run several linear regression models using weighted least squares. Determining whether these factors significantly impact a student's academic performance is the second biggest question of this study; determining what factors influence a student's health was the first.

In Section 1, we already discovered that the GPA data was heteroscedastic, so we needed to take caution in these analyses and use models that accurately represent this data. That's why we chose to use weighted least squares; WLS should account for the variance patterns of our data.

For our WLS regression, we ran 20 models to find which factors had statistical and practical significance. A table of ten of these models and their corresponding R^2 , p-values, and variance inflation factor values - for multicollinearity - are shown below. As shown through this table, we find that every model is statistically significant, with all the models having p-values less than $2.2e^{-16}$. However, we also see that increasing the complexity of the model (by adding more factors) also increases the R^2 , which shows practical significance to these models. Additionally, we find that VIF values remain fairly low until we start incorporating interaction of factors.

Given Table 6.1, we find that Model 8 explains 44% of the variance while keeping a relatively small VIF value, showing us statistical and practical significance in explaining the effects these factors have on GPA. While an adjusted R^2 value of 0.4448 is still fairly low, Model 8 appears to successfully explain the variance of GPA - better than most of the other models - and so, we find that the addition of all mental and physical health factors *do* significantly impact a student's GPA. Further analysis of this dataset needs to be done to adequately capture a better picture, but Model 8 isn't a terrible representation, especially considering this is for educational data, which is highly variable to begin with.

Additionally, we show the beta coefficients for Model 8 in Table 6.2. This model shows us that the greatest impact of a student's health on GPA is physical activity and sleep. With the greatest beta coefficient of 0.017, a student who does "moderate" levels of physical activity is predicted to have a greater GPA than a student who has "low" levels of physical activity. This is again true for "high" levels of physical activity, but it's much less than just simply "moderate". Looking at sleep quality, with a beta coefficient of -0.019, a student who has "average" sleep quality is actually predicted to have a lower GPA than a student who has "poor" sleep quality. This result is fascinating, and it's not an error. This could suggest that poor sleep quality isn't a deterrent for a student's GPA, and in fact, staying up late to complete assignments or class work may help students succeed academically. The other beta coefficients show similar stories, with anxiety appearing to be counterintuitive like sleep quality. For increased stress and depression, a

student's GPA decreases, and for better diet, a student's GPA increases. However, for increased anxiety, a student's GPA increases, which actually correlates to studies that show anxiety can be a powerful motivator. Thus, it appears greater anxiety will allow a student to excel academically; providing good motivation towards success.

What should be noted about these results is the factor of change that's shown in the effects plot and through the beta coefficients. We see that for any factor change, the predicted change in GPA is less than 0.02 across the board. This factor is incredibly small, and usually, a factor change this small corresponds to the difference between one letter grade (for *one* class) up or down (a B+ or A-). This just goes to say that while there does seem to be a change in GPA for each factor, it's relatively small, and it depends on the student whether this change is important to them (for scholarship or acceptance purposes). In sum, we do see significant effects these factors have on GPA, and Model 8 does explain nearly half of the variance, but they're relatively minor effects.

Model	Formula	Adj. R^2	Max VIF	p-val.
1	CGPA ~ Sleep + Depression	0.1414	1.0686	<2.2e-16
2	CGPA ~ Sleep + Anxiety	0.2465	1.0133	<2.2e-16
3	CGPA ~ Phys + Stress	0.2707	1.1383	<2.2e-16
4	CGPA ~ Diet + Phys	0.3005	1.1286	<2.2e-16
5	CGPA ~ Diet + Phys + Sleep	0.3306	1.2276	<2.2e-16
6	CGPA ~ Stress + Depression + Anxiety	0.2282	3.0697	<2.2e-16
7	CGPA ~ Phys*Diet	0.4141	1.5889	<2.2e-16
8	CGPA ~ Sum all factors	0.4448	2.4235	<2.2e-16
9	CGPA ~ Phys*Diet*Sleep	0.6120	76.4393	<2.2e-16
10	CGPA ~	0.1656	11.8951	<2.2e-16

	Stress*Depression			
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Table 6.1: Summarized table of ten different models using WLS regression. Model 8 appears to explain the greatest variance of GPA while keeping a low VIF value.

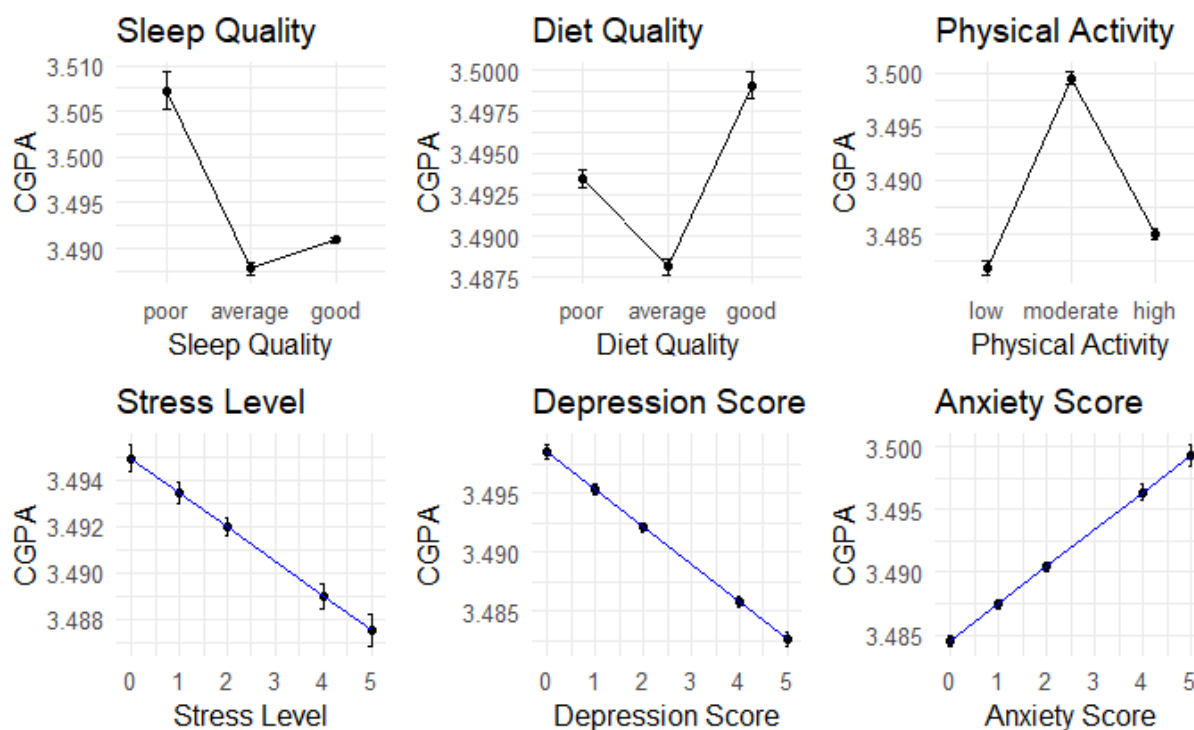


Figure 6.1: Effect plot of Model 8. Fascinating results showing each factor's effect on GPA. Sleep quality and anxiety appear counterintuitive, although could correspond to a student's effort to complete school work, which could allow for a greater GPA.

Factor	Beta Coefficients
(Intercept)	3.5037936
Sleep_Quality(average)	-0.0193825
Sleep_Quality(good)	-0.0162933
Diet_Qualityaverage	-0.0052986
Diet_Qualitygood	0.0056240

Physical_Activitymoderate	0.0176105
Physical_Activityhigh	0.0031947
Stress_Level	-0.0014892
Depression_Score	-0.0031715
Anxiety_Score	0.0029636

Table 6.2: Beta coefficients for each factor in Model 8. The coefficients are rather small, showing that changes in factors show a small predicted change in GPA.

Conclusion:

Our project aim is to identify trends between student health and habits, and academic performance. If we can better understand the factors that influence a student's mental and physical health, and whether they affect academic performance, we can implement strategies or programs in schools to help students overcome stressors and perform their best academically. For this project, we used two datasets from Kaggle: the *Student Mental Health Survey* dataset by Saint., and the *Student Health Data* dataset by Ziya in order to explore the relationship between students' mental and physical health.

In Section 1, we investigated the distribution of CGPA to see that it best follows a clustering model with six means and variances, so that it captures the data's heteroscedasticity. The distribution of resting heart rate best follows a normal distribution, with a mean of 70.10 bpm and a standard deviation of 9.54 bpm.

In Section 2, linear regression analysis shows that student's self-reported stress levels are significantly impacted by physical activity and health risk of a student. The second model using five factors accounts for 26% of the variance found in reported stress levels, a much greater improvement than the first model. This second model revealed that physical activity and health risk level are key predictors of how stressed a student *feels*. It supports the idea that students with lower physical activity levels and lower health risk scores report higher levels of perceived stress. Even though some predictors like mood and sleep quality might feel relevant they did not show a significant statistical relationship.

In Section 3, we used various hypothesis testing methods to examine different populations of the Student Mental Health Assessment dataset. These tests show some nearly significant results and noteworthy residuals but overall, they showed that there was little

significance between one group and the other. Using multiple linear regression and testing the interaction between *many* groups shows greater significance when it comes to mental and physical health, and academic performance. To see if sleep deprivation has a relationship with academic performance (CGPA) based on this dataset, a two-sample Welch t-test was completed. The outcome of the two-sample t-test ($t = -0.20408$, $p = 0.839$) concludes that there is no meaningful difference between GPA and sleep quality.

In Section 4, a one-way ANOVA test between physical activity and stress level resulted in nearly identical distributions between groups. This shows that just one factor cannot be attributed for the variance of stress level in students. Rather, a combination of factors and their interaction shows greater significance when it comes to mitigating stress levels.

In Section 5, using multiple linear regression from the first dataset, with categorical data such as sleep scores; it was not meaningful, since the numerical data was discrete, showing disconnected jumps in the graphs and low beta coefficient summary values in table 5.1. A better model was needed to determine how stress levels impact students. In this instance, the solution was to try an ordinal logistic regression model. Student Mental Health Regression Model analysis shows that students with medical courses are predicted to report higher stress levels. Additionally, students with chronic illness are predicted to report stress scores higher than zero. The students with medical courses are predicted to be far more likely to report higher stress levels.

Finally, for Section 6, we searched for the best linear model to predict GPA given mental and physical health. We found that a linear model that combines all the measured health factors - stress level, depression score, anxiety score, physical activity, diet quality, and sleep quality - appeared most successful, statistically significant, and accounts for 44% of the variance of GPA. These results show us that there's an inverse relationship between stress levels, depression score, and sleep quality, while there's a positive relationship between diet quality, physical activity, and anxiety score. These tests show the counterintuitive nature of sleep quality and anxiety's effects on GPA; both showing that essentially "worse" categories show a greater predicted GPA. This could suggest that anxiety can provide motivation and "poor" sleep could contribute to a greater GPA if students are sacrificing their sleep in order to complete school work. While there was significant effects of these factors on GPA, it was also found that these influences are rather minor, and the grade point change associated with a change in factor may only be important to students who need to keep GPAs above a certain level for scholarships or acceptance requirements.

For all of our analyses, we've shown that simple tests - Z-tests, chi-squared tests, and one-way ANOVA - all frequently lead to insignificant results for the given datasets. We obtained much more significant results through multiple regression models; through testing multiple factors and their interactions, we were able to find much more insightful results into what affects

students. For instance, using linear regression to test the effects of five different variables in Section 2 shows us that physical activity and health risk significantly impact a student's self reported stress level. This can be equated to say that a student's mental and physical health, and GPA, are not significantly affected by just one factor, but through interactions of *many* factors.

This makes it difficult to suggest remedies or programs that can better student's health or academic performance, because they are dependent on several factors. We suppose this is the difficult thing about this study, and researchers and institutions alike have been working on this exact problem for years: how to best support students so that they can succeed academically and have good health and wellness. What can be suggested to students is that every facet of their health is vitally important to them, and that in order to ensure the best academic performance, student's should prioritize keeping their health better than "poor" (unless it's sleep). Whether that's going to mindfulness meetings for stress and anxiety, joining groups to better their social support and thus, bettering their mental health overall, going for a run every once in a while, or choosing a well-balanced meal instead of vending machine junk, every one of these things, while they might seem minor, will help students in the long run. As for initiated programs in universities across the country, more analysis needs to be done to discern what groups of students truly will benefit from health and wellness programs, and whether those programs result in a large impact on GPA.

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